

2.10.11

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Question:

A unit vector that is coplanar with $\hat{i} + \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} + \hat{k}$ and perpendicular to $\hat{i} + \hat{j} + \hat{k}$ is----

Solution:

Let the two vectors lie on plane **P**

as origin also lies on the plane

The Equation of the plane is $\mathbf{r} \cdot \mathbf{n}^T = 0$

$$\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \cdot \mathbf{n}^T = 0 \quad (0.1)$$

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \mathbf{n}^T = 0 \quad (0.2)$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix} \cdot \mathbf{n} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.3)$$

$$(0.4)$$

The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 0 \end{array} \right) \quad (0.5)$$

(0.6)

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_1 \rightarrow R_1 - R_2$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right) \quad (0.7)$$

we get

$$\mathbf{n} = \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix} \quad (0.8)$$

Let the unit vector be **A**

$$\mathbf{B} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (0.9)$$

$$(\mathbf{n} \quad \mathbf{B})^T \cdot \mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.10)$$

$$\begin{pmatrix} -3 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \cdot \mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.11)$$

The augmented matrix is

$$\left(\begin{array}{ccc|c} -3 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right) \quad (0.12)$$

$$R_1 \longrightarrow R_1 - R_2 \text{ and } R_2 \longrightarrow R_2 + R_1/4$$

$$\left(\begin{array}{ccc|c} -4 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right) \quad (0.13)$$

$$\mathbf{A} = \begin{pmatrix} 0 \\ b \\ -b \end{pmatrix} \quad (0.14)$$

As $\mathbf{A}$ is a unit vector

$$\therefore \mathbf{A} = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \quad (0.15)$$

Vectors and Plane spanned by u and v

